



* Practical - 3 *

- * Authentication, Validation and Toast Develop an authentication application that asks user to enter email id and password. Log in Button should remain disabled until user enter email id in valid format. In case of successful authentication, user should be redirected to another activity which contains welcome message along with email id of authenticated user, otherwise appropriate error message should be displayed.

* activity_main.xml

```
<RelativeLayout xmlns:android="
    http://schemas.android.com/apk/res/
    android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
```

```
<EditText
```

```
    android:id="@+id/email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email"
    android:inputType="textEmailAddress"/>
```



<EditText

```
android:id="@+id/password"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:hint="Password"  
android:layout_below="@id/email"  
android:layout_marginTop="10dp"  
android:inputType="textPassword"/>
```

<Button

```
android:id="@+id/loginButton"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:text="Login"  
android:layout_below="@id/password"  
android:layout_marginTop="20dp"  
android:enabled="false"/>
```

</RelativeLayout>

* MainActivity.java

```
package com.example.emailemailvalidationlogin;
```

```
import android.os.Bundle;  
import android.text.Editable;  
import android.text.TextWatcher;  
import android.util.Patterns;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;
```




```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
    EditText email, password;
    Button loginButton;
```

⑨Override

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
    email = findViewById(R.id.email);
    password = findViewById(R.id.password);
    loginButton = findViewById(R.id.loginButton);
```

```
    TextWatcher textWatcher = new TextWatcher() {
```

⑨Override

```
    public void beforeTextChanged(CharSequence s,
        int start, int count, int after) {}
    }
```

⑨Override

```
    public void onTextChanged(CharSequence s,
        int start, int count, int after) {}
    }
```

⑨Override

```
    public void afterTextChanged(CharSequence s) {}
    }
};
```



```
email.addTextChangedListener(CtextWatcher);
password.addTextChangedListener(CtextWatcher);
```

```
loginButton.setOnClickListener(Cnew View.OnClickListener){
    @Override
```

```
    public void onClick(CView v){
```

```
        Toast.makeText(CMainActivity.this, "Login Successful",
            Toast.LENGTH_SHORT).show();
```

```
    }
```

```
}
```

```
}
```

```
private void validateInputs(){
```

```
    String emailInput = email.getText().toString().trim();
```

```
    String passwordInput = password.getText().toString().trim();
```

```
    if (Patterns.EMAIL_ADDRESS.matcher(emailInput).
```

```
        matches() && loginButton.setEnabled(true);
```

```
    } else {
```

```
        loginButton.setEnabled(false);
```

```
    }
```

```
}
```

```
}
```


* Output *